

Hanoi University of Science and Technology
IT5425: Data Management and Visualization

Olist InsightHub: From Data Exploration to Strategic Storytelling

Capstone Project Report

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Abstract

This report presents **Olist InsightHub**, a data visualization project based on the Brazilian Olist e-commerce dataset. The project follows a hybrid workflow: Python is used for data preparation and exploratory data analysis, while Tableau is used for the final interactive dashboard. The analysis focuses on three business questions: where revenue comes from, where fulfillment problems occur, and how delivery performance affects customer review scores.

1 Project Overview

1.1 Business Questions

The project is designed around three questions:

1. Where does Olist's revenue come from?
2. Where are the main logistics and delivery issues?
3. How does delivery performance affect customer satisfaction?

1.2 Workflow

The project uses a hybrid approach. Python, Pandas, and Plotly are used to clean the data, create derived variables, validate joins, and explore patterns. Tableau is used to build the explanatory dashboard, where users can interact with filters and drill into the business story.

2 Data Preparation

2.1 Dataset Structure

The raw Olist dataset contains nine related CSV files: `orders`, `order_items`, `payments`, `reviews`, `customers`, `sellers`, `products`, `product_category_translation`, and `geolocation`. These tables have different levels of granularity. For example, `orders` is at order level, `order_items` is at item level, and `payments` may contain multiple rows for one order.

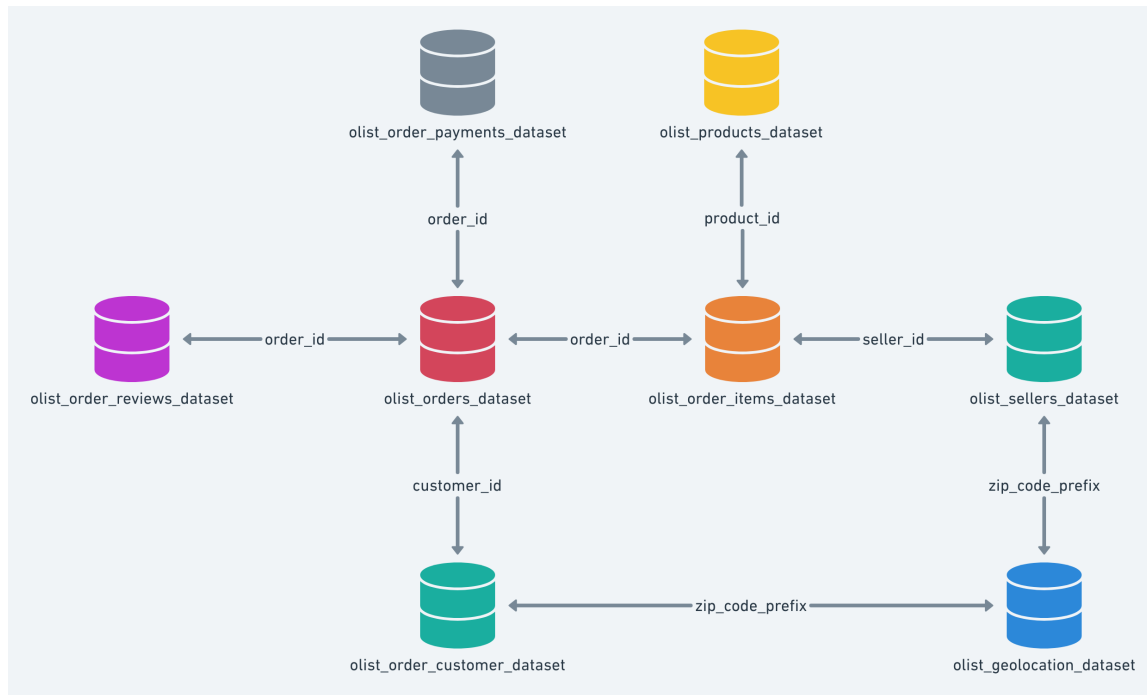


Figure 1: Relationship diagram of the Olist dataset

2.2 Cleaning Strategy

The main data preparation risk is double counting. If item-level, payment-level, and review-level tables are joined directly, revenue and review metrics can be duplicated. Therefore, payments and reviews were first aggregated to **order_id**. The geolocation table was also compressed by **zip_code_prefix**, reducing the size of the spatial lookup table and improving performance.

The final cleaned data model contains two main analytical tables:

Table 1: Main cleaned analytical tables

Table	Purpose
<code>mart_order_level.csv</code>	Order-level analysis: delivery, delay, payment, review, customer location.
<code>mart_item_level.csv</code>	Item-level analysis: revenue, product category, seller, freight, distance.

2.3 Key Derived Variables

Several derived variables were created to support later analysis:

- **shipping_days**: time between purchase and delivery.
- **delivery_delay_days**: difference between actual and estimated delivery date.
- **is_delayed**: whether an order arrived later than expected.

- `purchase_year_month`: monthly time period for trend analysis.
- `freight_ratio`: freight value divided by item price.
- `seller_customer_distance_km`: approximate seller-customer distance.

3 Exploratory Data Analysis

3.1 EDA Direction

The EDA phase is organized as a business funnel:

Demand → Fulfillment → Satisfaction

This structure keeps the analysis focused. Demand analysis identifies where revenue comes from. Fulfillment analysis checks delivery performance and logistics risk. Satisfaction analysis connects delivery performance with customer review scores.

3.2 Key Metrics

Table 2: Overall EDA metrics

Metric	Value
GMV	13,591,643.70
Orders	99,441
Delivered Orders	96,478
AOV	136.68
On-time Rate	91.89%
Delayed Rate	8.11%
Average Review Score	4.09
Median Shipping Days	10.22

3.3 Initial Findings

The EDA produced several initial business observations:

- The top revenue category is `health_beauty`, with about 1.26 million GMV.
- SP is the largest customer state, with about 5.20 million GMV.
- The overall delayed rate is about 8.11%.
- Review scores decrease clearly when delivery delay becomes severe.
- Distance and shipping time show a moderate positive relationship in the EDA sample.

3.4 EDA Figures and Insights

The following figures are static PNG exports prepared from the Phase 2 EDA outputs. They are included to support the business interpretation before building the final Tableau dashboard.

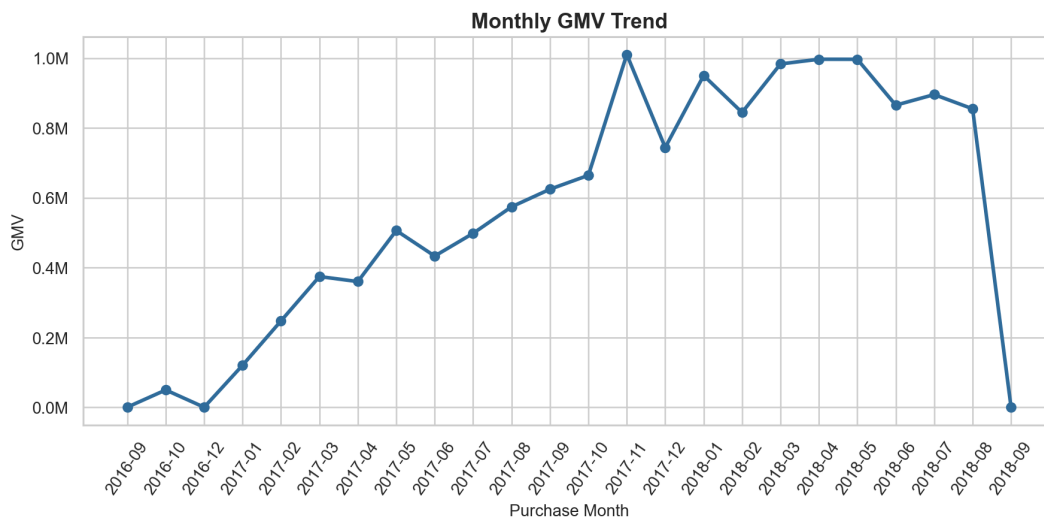


Figure 2: Monthly GMV Trend from Python EDA

Insight. Monthly GMV grows strongly from 2017 onward, with a visible peak around the late-2017 shopping season and sustained high revenue in several 2018 months. This suggests that the dashboard should keep time filtering available, because revenue patterns are not constant across months.

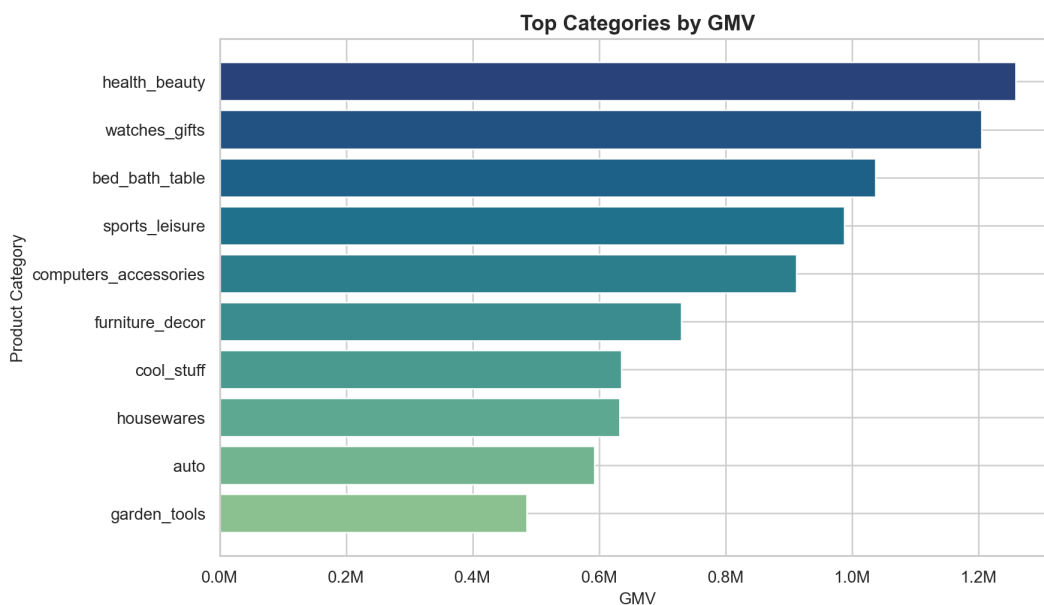


Figure 3: Top Categories by GMV from Python EDA

Insight. Revenue is concentrated in a small group of categories, led by `health_beauty`, `watches_gifts`, and `bed_bath_table`. These categories should be highlighted in the

Sales tab because they explain a large part of Olist's GMV.

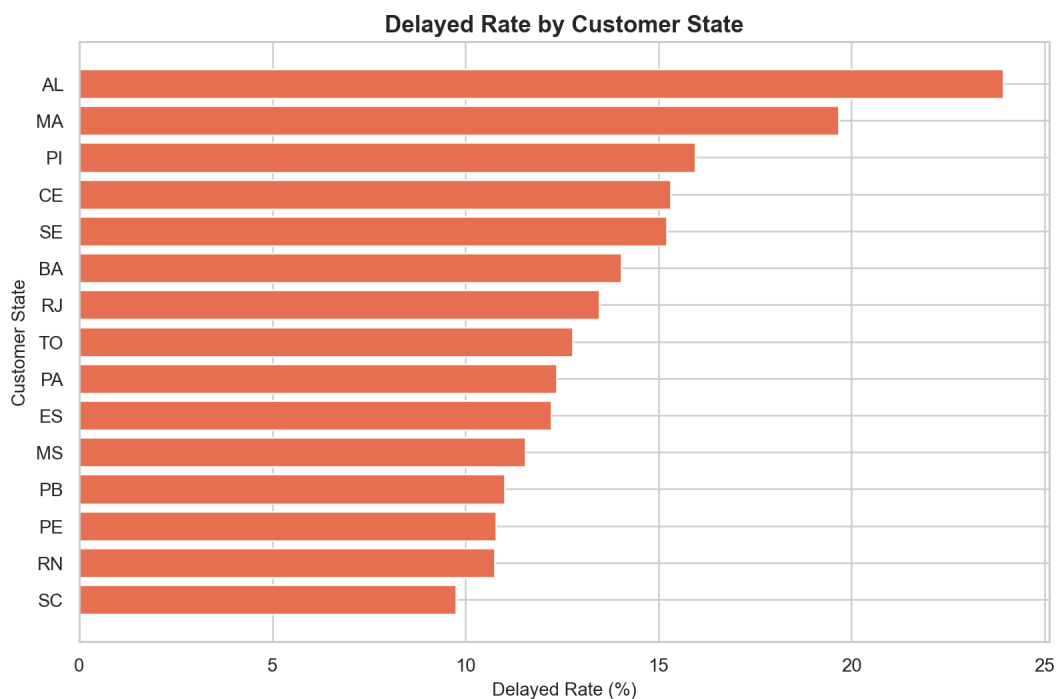


Figure 4: Delayed Rate by State from Python EDA

Insight. Delivery delay is geographically uneven. Some states show much higher delayed rates than the platform average, meaning that logistics performance should be analyzed regionally instead of only through one global KPI.

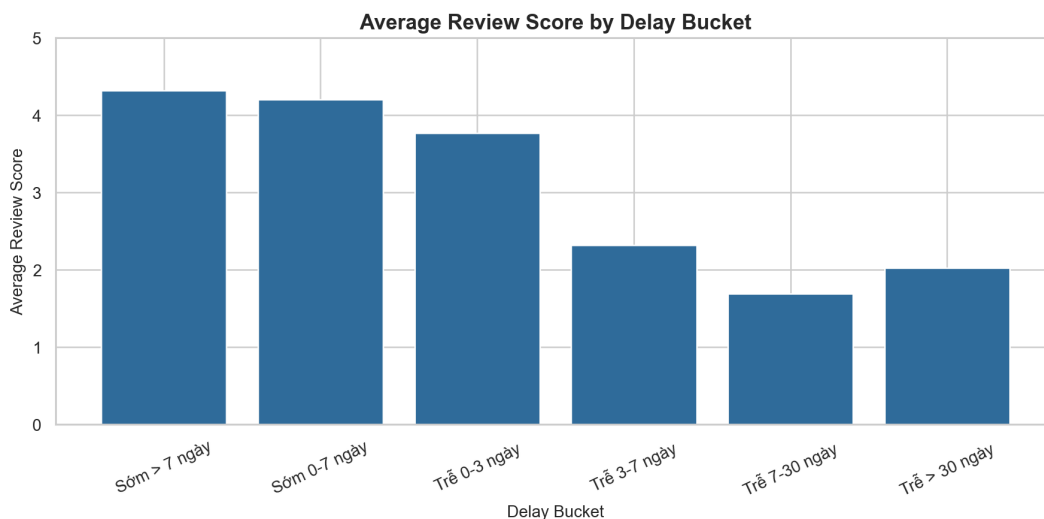


Figure 5: Average Review Score by Delay Bucket from Python EDA

Insight. Review scores decline sharply as delivery delay becomes more severe. Orders delivered early or on time receive average scores above 4, while orders delayed by more than a week receive much lower ratings. This supports the main storytelling link between fulfillment quality and customer satisfaction.

4 Tableau Dashboard Design

4.1 Dashboard Structure

The Tableau dashboard is designed with three main tabs:

1. **Sales / Demand Overview**
2. **Logistics / Fulfillment Overview**
3. **Customer Review / Satisfaction Overview**

Together, the three tabs form a business narrative sequence:

Demand → Fulfillment → Satisfaction

The first tab identifies where revenue comes from. The second tab evaluates delivery and fulfillment performance. The third tab connects fulfillment performance with customer review outcomes. Therefore, the dashboard is not only a collection of separate charts, but an interactive analytical flow that allows viewers to move from revenue performance to operational performance and finally to customer satisfaction.

4.2 Tab 1: Sales / Demand Overview

This tab focuses on the question: **Where does revenue come from?**

Meaning of the tab. The Sales / Demand Overview tab provides a high-level view of business performance. It helps users understand the overall scale of sales, how GMV changes over time, which product categories contribute the most revenue, and which customer states generate the largest demand. This tab is mainly designed for business, sales, and category-management viewers who need to quickly identify the main revenue drivers.

Main data source. This tab uses `mart_item_level.csv`, because GMV, item price, freight value, product category, and item-level revenue are naturally analyzed at the item level.

- **KPI cards:**
 - **GMV:** total merchandise value calculated from item prices.
 - **Orders:** number of distinct orders.
 - **AOV:** average order value.
 - **Freight Ratio:** freight cost as a percentage of item revenue.
- **Main charts:**
 - **Monthly GMV Trend:** shows how GMV changes over time.

- **Top Categories by GMV:** shows the top product categories contributing to revenue.
- **GMV by Customer State:** shows the customer states that contribute the most GMV.
- **Filters and user interaction:**
 - **Purchase Date filter:** allows users to select a specific time range. When the selected period changes, the KPI cards and all charts are recalculated for that period.
 - **Customer State filter:** allows users to focus on one or more customer states. This helps viewers compare the revenue pattern of specific states against the overall market.

How users can use this tab. A viewer can first look at the KPI cards to understand the overall sales scale. Then, the monthly trend shows whether GMV is increasing, decreasing, or fluctuating over time. The category chart identifies the main product groups driving revenue, while the state chart shows whether demand is geographically concentrated. For example, a user can select a particular customer state and immediately see how the GMV trend and top categories change for that state.

Design rationale. The tab starts with KPI cards because they provide a fast executive summary. The line chart supports temporal analysis, while horizontal bar charts make category and geographic comparisons easier to read. The filters turn the tab from a static summary into an exploratory sales dashboard.

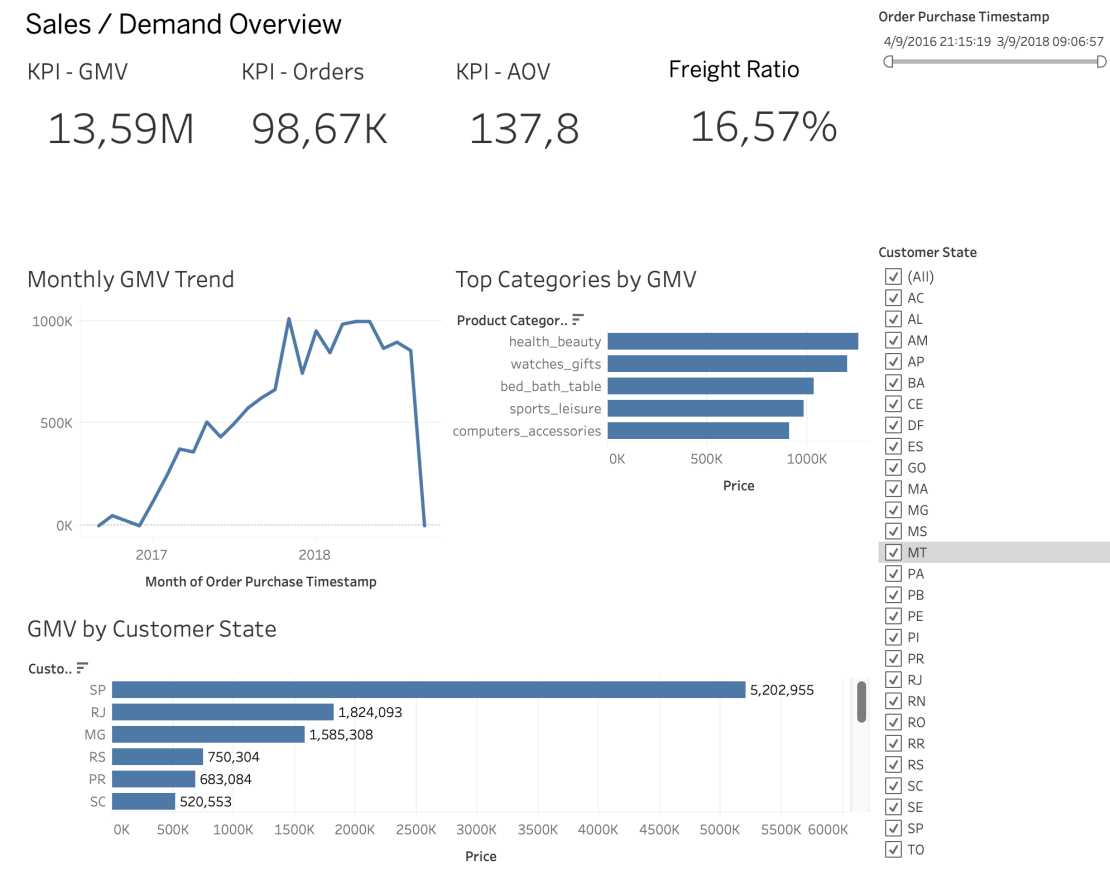


Figure 6: Tableau Dashboard Tab 1: Sales / Demand Overview

4.3 Tab 2: Logistics / Fulfillment Overview

This tab focuses on the question: **Where are delivery problems concentrated?**

Meaning of the tab. The Logistics / Fulfillment Overview tab evaluates delivery performance after demand has been identified. It helps users understand how many orders are delivered on time, how often delays occur, how long shipping usually takes, and which customer states suffer from heavier delay problems. This tab is mainly designed for operations, logistics, and fulfillment viewers.

Main data source. This tab uses `mart_order_level.csv`, because delivery status, shipping days, delay days, and customer state are analyzed at the order level.

- **KPI cards:**

- **Total Orders:** number of distinct orders included in the selected view.
- **Delayed Rate:** percentage of orders delivered later than the estimated delivery date.
- **On-time Rate:** percentage of orders delivered on time or earlier than expected.
- **Median Shipping Days:** median number of days required to deliver an order.

- **Main charts:**
 - **Delayed Rate by Customer State:** shows the customer states with the highest percentage of delayed orders.
 - **Shipping Days Distribution:** shows the distribution of delivery duration across orders.
 - **Average Delay Burden by Customer State:** shows the average delay impact by state. In this metric, non-delayed orders are counted as zero delay days.
- **Filters and user interaction:**
 - **Purchase Date filter:** allows users to analyze delivery performance within a selected time range. This is useful for checking whether logistics performance changes over time.
 - **Customer State filter:** allows users to focus on one or more states and compare their delivery performance with the overall market.
 - **Is Delivered filter:** allows users to include all orders or focus only on completed deliveries. This is important because shipping days and delivery delay are easier to interpret when the order has actually been delivered.

How users can use this tab. A viewer can start from the KPI cards to understand the overall fulfillment condition. The delayed-rate chart then shows where delivery problems are most frequent. The shipping distribution explains the normal delivery-time pattern, while the delay-burden chart highlights states where the average delay impact is more severe. For example, a logistics analyst can select a specific customer state and check whether that state has both a high delayed rate and a high delay burden.

Design rationale. This tab separates delay frequency from delay severity. *Delayed Rate* answers how often delays happen, while *Average Delay Burden* answers how severe the delay impact is on average. The shipping distribution provides context by showing the overall pattern of delivery duration. The *Is Delivered* filter gives users control over whether they want to analyze all orders or focus on completed deliveries.

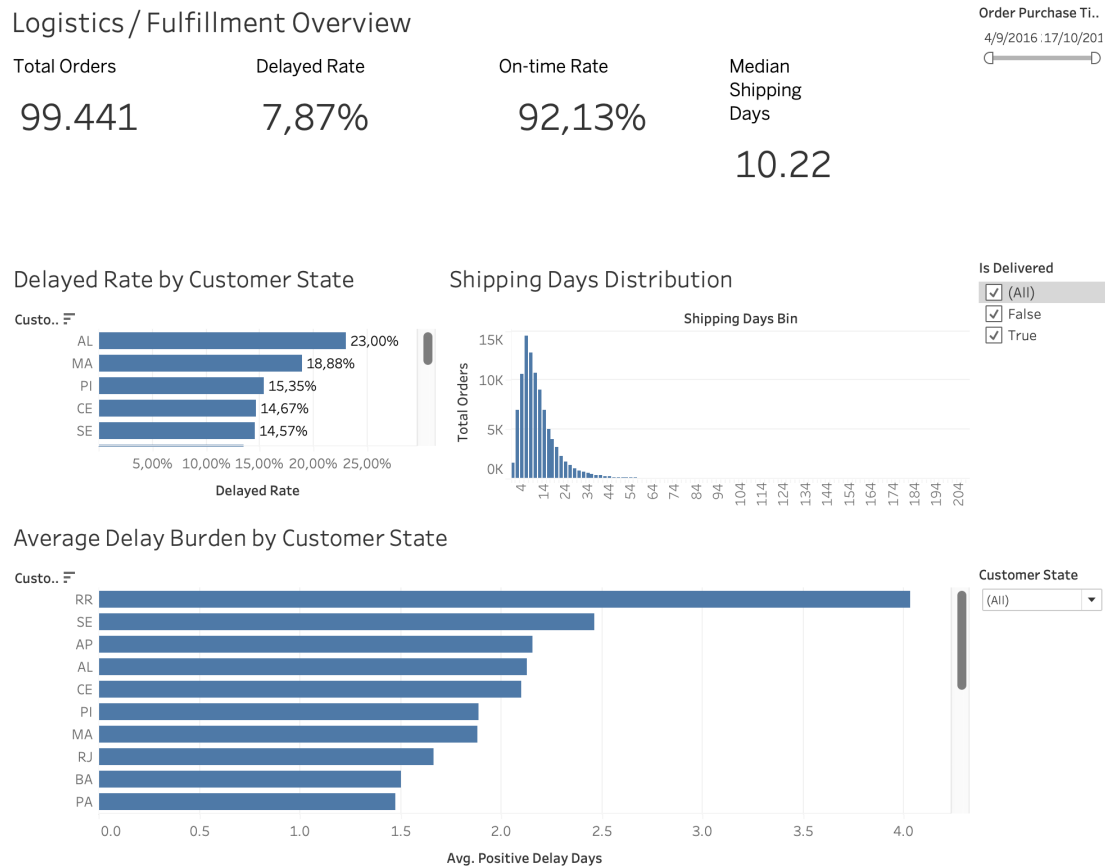


Figure 7: Tableau Dashboard Tab 2: Logistics / Fulfillment Overview

4.4 Tab 3: Customer Review / Satisfaction Overview

This tab focuses on the question: **How does fulfillment affect customer satisfaction?**

Meaning of the tab. The Customer Review / Satisfaction Overview tab connects operational performance with customer perception. It helps users understand the overall review distribution, whether delayed orders receive lower review scores, and whether longer delays lead to worse customer satisfaction. This tab is mainly designed for customer experience, service quality, and management viewers.

Main data source. This tab uses `mart_order_level.csv`, because review scores and delay indicators are analyzed at the order level.

- **KPI cards:**

- **Average Review Score:** average customer review score.
- **Low Review Rate:** percentage of reviewed orders with low scores, defined as 1 or 2 stars.
- **Positive Review Rate:** percentage of reviewed orders with positive scores, defined as 4 or 5 stars.
- **Reviewed Orders:** number of orders that have review information.

- **Main charts:**

- **Review Score Distribution:** shows how reviews are distributed from 1 to 5 stars.
 - **Average Review Score by Delay Status:** compares review scores between on-time or early orders and delayed orders.
 - **Review Score by Delay Bucket:** shows how average review score changes as delivery delay becomes more severe.
- **Filters and user interaction:**
 - **Purchase Date filter:** allows users to examine customer satisfaction within a selected time period.
 - **Customer State filter:** allows users to compare customer satisfaction patterns across different states.
 - **Is Delivered filter:** allows users to include all orders or focus only on delivered orders when evaluating the relationship between delivery delay and review score.

How users can use this tab. A viewer can first check the KPI cards to understand the overall satisfaction level. The review distribution then shows whether most customers leave positive or negative ratings. Next, the delay-status chart directly compares on-time or early orders with delayed orders. Finally, the delay-bucket chart shows whether customer ratings decline as delays become longer. For example, a manager can select a specific customer state and examine whether delay-related dissatisfaction is stronger in that region.

Design rationale. This tab is the strongest storytelling component of the dashboard because it connects fulfillment quality to customer satisfaction. The design moves from a general satisfaction baseline to a direct comparison between delayed and non-delayed orders, and finally to a more detailed delay-severity analysis. This supports the interpretation that logistics performance is not only an operational issue but also a customer-experience issue.

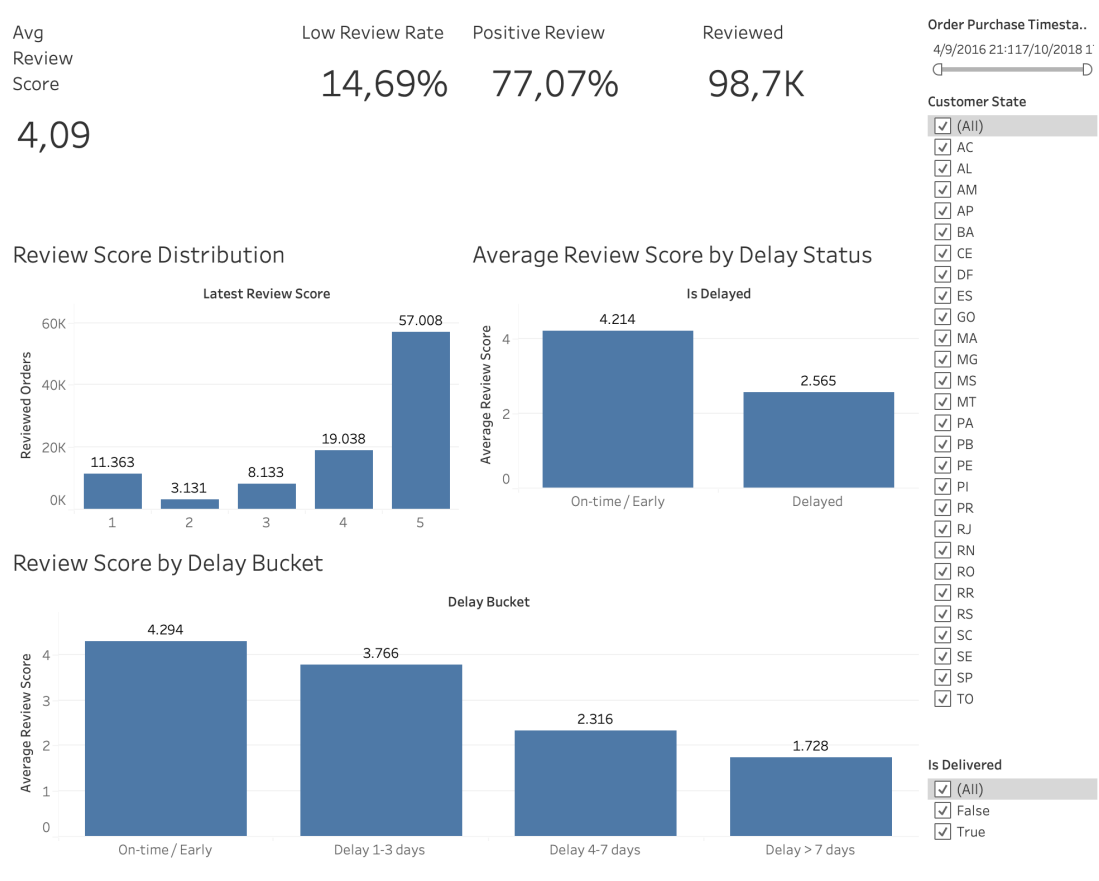


Figure 8: Tableau Dashboard Tab 3: Customer Review / Satisfaction Overview

4.5 Summary of Dashboard Narrative

The three tabs work together as an integrated dashboard story. The *Sales / Demand Overview* tab identifies where business value is generated. The *Logistics / Fulfillment Overview* tab evaluates whether the order fulfillment process can support that demand reliably. The *Customer Review / Satisfaction Overview* tab shows how delivery performance is reflected in customer reviews. With interactive filters for time, geography, and delivery status, users can move from a general overview to more specific business questions without changing the dashboard structure.

5 Technique Application

5.1 Chapter 1: Overview of Data Visualization

Techniques / principles applied. The project uses both exploratory visualization and explanatory visualization. Python supports exploration, while Tableau supports dashboard-based communication.

How applied in the dashboard. The Tableau dashboard is structured around three business questions: revenue, fulfillment, and satisfaction. The three dashboard tabs follow the sequence *Demand* → *Fulfillment* → *Satisfaction*. This structure allows users to first

understand revenue patterns, then inspect logistics performance, and finally evaluate how delivery performance is related to customer review outcomes.

Notes / adjustments. The report focuses on relevant views instead of including every possible chart from the dataset.

5.2 Chapter 2: Visual Models and Encoding

Techniques / principles applied. The project uses nominal, ordinal, temporal, geographic, and quantitative variables. The main visual channels are position, length, and text labels.

How applied in the dashboard. GMV, Orders, AOV, Freight Ratio, Delayed Rate, On-time Rate, Shipping Days, Review Score, and Review Rate are quantitative variables. Product Category and Customer State are nominal variables. Review Score is treated as an ordinal satisfaction measure. Bar charts use position and length for category, state, delay, and review comparisons. The line chart encodes time through the Monthly GMV Trend. Geographic information is represented through the Customer State dimension rather than through a map view in the final Tableau dashboard.

Notes / adjustments. Position and length are preferred for core comparisons because they are easier to read than angle or area.

5.3 Chapter 3: Graphical Perception

Techniques / principles applied. The dashboard uses pre-attentive processing, Gestalt grouping, and magnitude estimation.

How applied in the dashboard. KPI cards are placed at the top of each tab so that viewers can quickly capture the most important indicators before reading the detailed charts. Related charts are grouped by topic: sales, logistics, and customer satisfaction. The dashboard uses consistent bar and line encodings, while labels are added to important KPI cards and selected bar charts to reduce interpretation effort.

Notes / adjustments. Color is used sparingly to avoid visual overload. The dashboard mainly relies on layout, position, length, and labels rather than heavy color encoding.

5.4 Chapter 4: Visualization for Multi-dimensional Data

Techniques / principles applied. The dashboard covers amounts, trends, distributions, proportions, and relationships between operational and customer-experience variables.

How applied in the dashboard. Monthly GMV Trend uses a line chart to show revenue over time. Top Categories by GMV and GMV by Customer State use horizontal bar charts to compare revenue across categories and states. Shipping Days Distribution uses a histogram to show the distribution of delivery duration. Delayed Rate by Customer State and Average Delay Burden by Customer State compare logistics performance across

geographic groups. Review Score Distribution shows the distribution of customer ratings, while Average Review Score by Delay Status and Review Score by Delay Bucket compare satisfaction across delivery-delay groups.

Notes / adjustments. Uncertainty is not shown as a separate view. Instead, medians and distributions are used to avoid relying only on averages.

5.5 Chapter 5: Visualization for Graphs

Techniques / principles applied. Graph visualization is not a major component of this project.

How applied in the dashboard. The Olist data is relational, but the dashboard does not model it as a node-link network. The relationship diagram in Figure 1 is used for data modeling, not as a dashboard chart.

Notes / adjustments. Graph charts were excluded because they do not directly support the main business questions.

5.6 Chapter 6: Principles of Figure Design

Techniques / principles applied. The design applies proportional ink, clutter reduction, controlled color, clear titles, and avoids unnecessary 3D charts.

How applied in the dashboard. Bar charts start from zero to preserve proportional ink. Horizontal bar charts are used for state and category comparisons because they are easier to read when labels are long. Top-N filtering is used in several views, such as Top Categories by GMV, GMV by Customer State, Delayed Rate by Customer State, and Average Delay Burden by Customer State, to reduce clutter and keep the dashboard focused. The dashboard is divided into three tabs so that each page focuses on one business theme.

Notes / adjustments. No 3D charts are used because they would reduce readability without adding analytical value.

5.7 Chapter 7: Map Visualization

Techniques / principles applied. Map visualization is relevant to this dataset because the Olist data contains geographic information such as customer state, seller location, and zip-code-prefix-based location attributes. However, map visualization is not directly implemented in the final Tableau dashboard.

How applied in the dashboard. Instead of using a map view, the final dashboard represents geographic patterns through **Customer State** bar charts. In the Sales tab, *GMV by Customer State* shows geographic revenue concentration. In the Logistics tab, *Delayed Rate by Customer State* and *Average Delay Burden by Customer State* compare delivery problems across states.

Notes / adjustments. Map views were considered but excluded from the final dashboard to keep the analysis clearer and more focused. Since the main business questions

require comparing GMV, delayed rate, and delay burden across states, horizontal bar charts provide more precise value comparison than a map. In addition, the available coordinates are based on zip code prefixes, so they are more suitable for regional-level analysis than exact address-level analysis.

5.8 Chapter 8: Interactive Visualization

Techniques / principles applied. The dashboard uses filtering and view transformation to support interactive exploration.

How applied in the dashboard. The Sales tab uses Purchase Date and Customer State filters. The Logistics tab uses Purchase Date, Customer State, and Is Delivered filters. The Customer Review tab also uses Purchase Date, Customer State, and Is Delivered filters. These filters allow users to change the time period, focus on specific customer states, and decide whether to include all orders or focus on delivered orders. When a filter is adjusted, the KPI cards and charts in the corresponding tab are recalculated.

Notes / adjustments. Animation is not used because filtering is more useful for this business analysis. The interaction design is kept simple so that viewers can explore the dashboards without needing complex navigation.

5.9 Chapter 9: Storytelling with Data

Techniques / principles applied. The dashboard uses narrative flow, focused message, and interaction in storytelling.

How applied in the dashboard. The dashboard story follows a simple sequence: Sales shows where revenue comes from, Logistics shows where delivery problems occur, and Customer Review shows how delivery delay is associated with customer satisfaction. This structure helps viewers move from business performance to operational performance and then to customer experience.

Notes / adjustments. Dashboard screenshots are added to document the final Tableau workbook and support the report discussion.

6 Conclusion

The project builds a clean analytical data model and a business-focused visualization plan for the Olist dataset. Python supports data preparation and EDA, while Tableau will provide the final interactive dashboard. The expected dashboard story connects revenue concentration, logistics performance, and customer satisfaction into one coherent business narrative.